Zaen Quaisar

(203) 628 5690 | quaisar.z@northeastern.edu | linkedin.com/in/zaen-quaisar | zaenquaisar.github.io/portfolio/

EDUCATION

Northeastern University, College of Engineering **Boston, MA**
Master of Science in Energy Systems – GPA: 3.92 *Expected December 2026*
Bachelor of Science in Mechanical Engineering, Minor in Sustainable Energy Systems – GPA: 3.85 *May 2026*

SKILLS AND INTERESTS

Design and Analysis: SolidWorks, MATLAB, C++, KiCad, OnShape, OCalc Pro, GIS
Prototyping and Fabrication: 3D Printing, Arduino, ESP32, Soldering, Power Tools, Hand Tools
Interests: Woodworking, DJing, Cooking, Skiing, Reading, Tennis, Drumming, Gaming

PROFESSIONAL EXPERIENCE

SharkNinja **Needham, MA**
R&D Engineering Co-Op, Home Environment *Jan 2025 – June 2025*

- Designed and executed experiments using specialized airflow test equipment including balometers, venturi rigs, airwatts measurement systems, and large anemometer arrays to optimize performance metrics
- Analyzed experimental data in MATLAB and Python to identify design levers affecting airflow efficiency and acoustic performance, providing quantitative feedback to inform future designs
- Designed and fabricated 80/20 aluminum test rig with integrated load cell and Arduino-based DAQ system to quantify wind load metrics, measuring instantaneous and time-varying forces with real-time LCD display
- Developed custom Arduino-based motor controller with PWM capabilities to characterize motor performance and conduct thermal analysis across operating conditions

ControlPoint Technologies Inc. **Rockland, MA**
Mechanical Engineering Co-Op *Jan 2024 – June 2024*

- Designed and enhanced electric power distribution lines across Massachusetts using National Grid GIS and OCalc Pro to optimize electrical flow, improve reliability, and reduce power loss
- Collaborated with engineers, project managers, and field technicians to implement design improvements and ensure compliance with industry standards and safety regulations
- Developed clear and accurate packages of supporting documentation including construction sketches, material lists, environmental assessments, and Digsafe tickets using Microsoft Word, Visio, and Excel
- Conducted field surveys and performed structural/load analysis on utility poles to ensure NESC compliance and optimize placement

LEADERSHIP AND ACTIVITIES

Vice President of External Relations and Logistics **Boston, MA**
Northeastern University Energy Systems Society *Spring 2026 - Present*

- Serve as primary liaison between NU-ESS and industry/academic partners across the energy sector, managing outreach and ongoing relationships with corporate and research organizations
- Recruit and coordinate 5–6 industry panels for NU-ESS's annual energy conference (~300 attendees), owning speaker outreach and communication from first contact through event day
- Manage full event logistics for NU-ESS programming including scheduling, room bookings, vendor sourcing, and supply/catering coordination, all in compliance with university CSI policies

ENGINEERING PROJECTS

Dual Axis Solar Tracker with Portable Power System **Boston, MA**
Independent *Spring 2026 - Present*

- Designed and built a dual-axis solar tracking system (ESP32-based) achieving accuracy within 2° for under $700
- Engineered closed-loop motor control using worm gears and combining a quadrature-encoded DC motor (azimuth) and microstepped stepper motor (elevation), with dual INA3221 sensors for power monitoring
- Embedded battery storage system with BMS and USB-C power delivery (5V/9V/12V/20V)
- Architected FSM-based firmware (C++/Arduino) across modular subsystems like motor control, power management, sun-position calculation, and safety, with a serial interface for real-time control